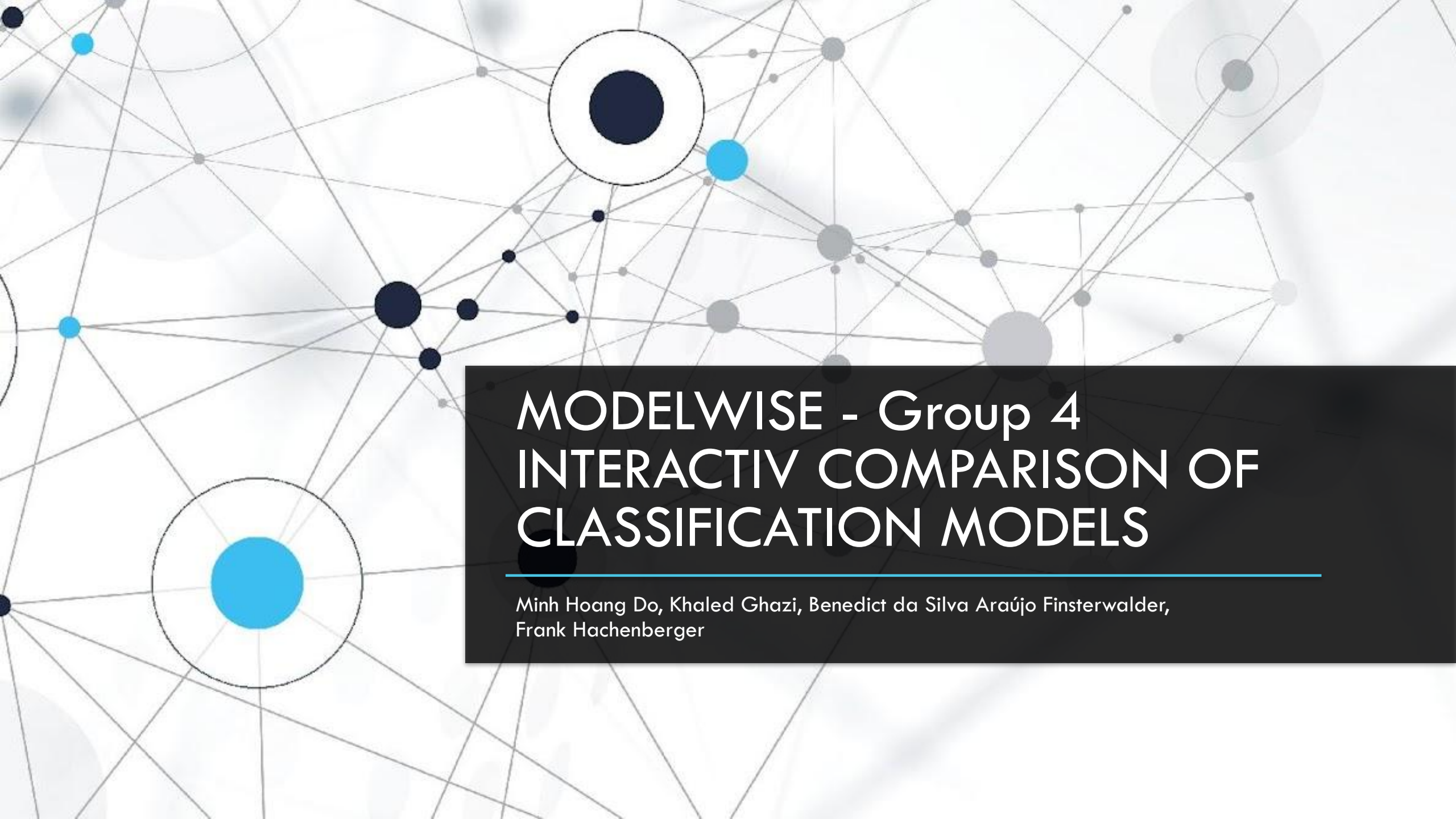


The background of the slide is a complex, abstract network diagram. It consists of numerous nodes of varying sizes and colors (dark blue, light blue, and grey) connected by thin, light grey lines. Some nodes are highlighted with larger, concentric circles. The overall aesthetic is technical and interconnected.

MODELWISE - Group 4 INTERACTIV COMPARISON OF CLASSIFICATION MODELS

Minh Hoang Do, Khaled Ghazi, Benedict da Silva Araújo Finsterwalder,
Frank Hachenberger

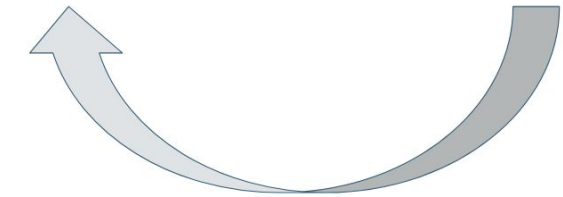
Project

- What was the Problem?
 - Understanding the data
 - Training a (good) classification model
 - Understanding the model (and data) → XAI
- Why is it important to address this problem?
 - Because we are Software Engineers and Data Scientists (to be)
 - To understand behaviour and predictions (no black-box)
 - And visualize our results

Milestones

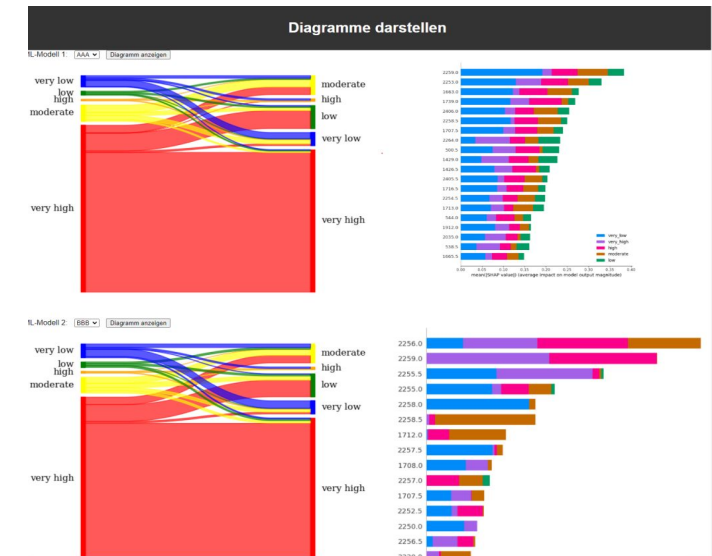
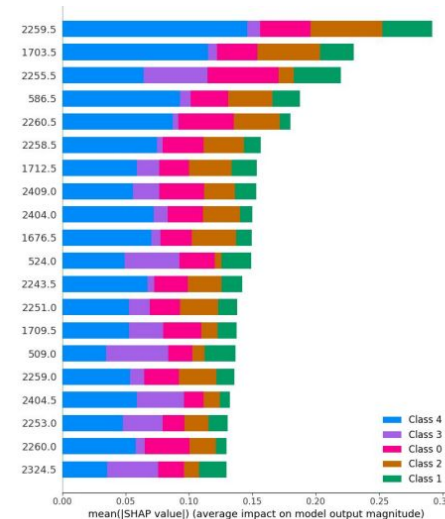
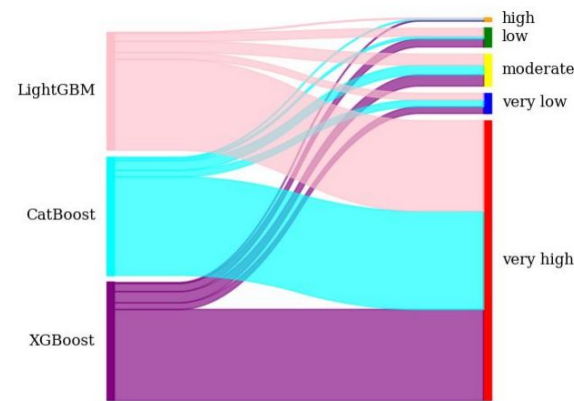
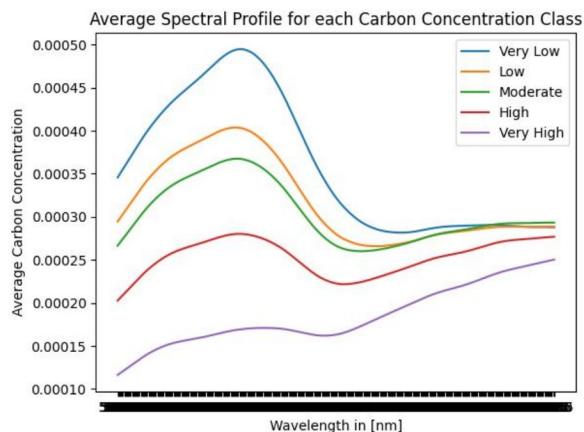
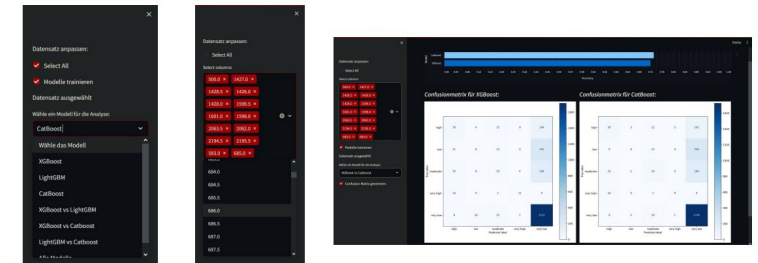
1. Exploration of Data (understanding data)
2. Exploration of possible Visualization methods (sankey, confusion,...)
3. Add Shapley-Values
4. Presentation of XAI concept
5. low/high fidelity design of application/visualization

User can change data(set) → See changes in model



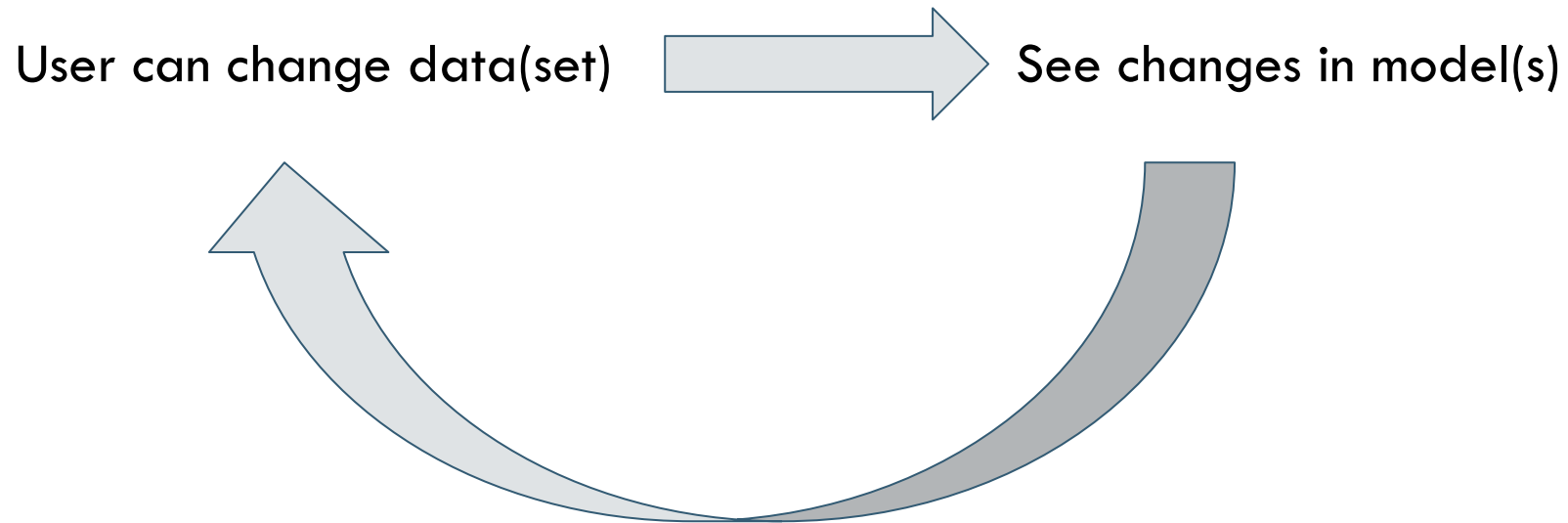
Streamlit

is an open-source Python framework for machine learning and data science



XAI Strategy

- User chooses features
- User chooses model or models to compare



XAI Implementation

4 different visualisations and 2 different comparisons

Visualisation: accuracy, sankey, shapley and confusion

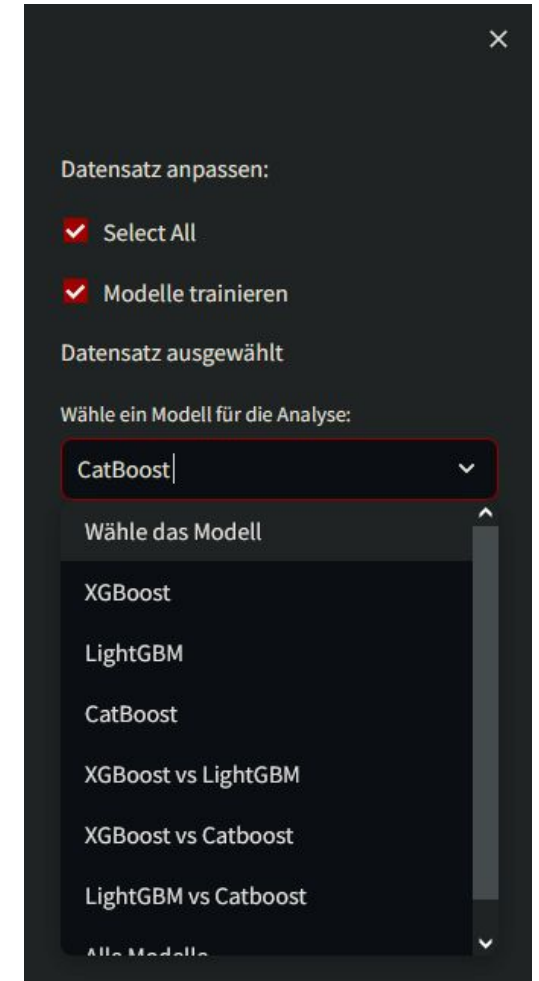
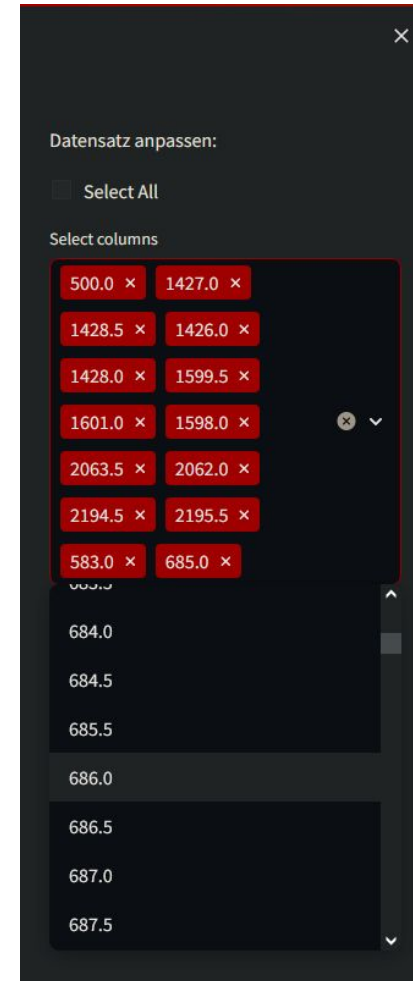
Comparisons:

- 2 models or 3 models on same dataset
- 1 model on two datasets

Using: Streamlit

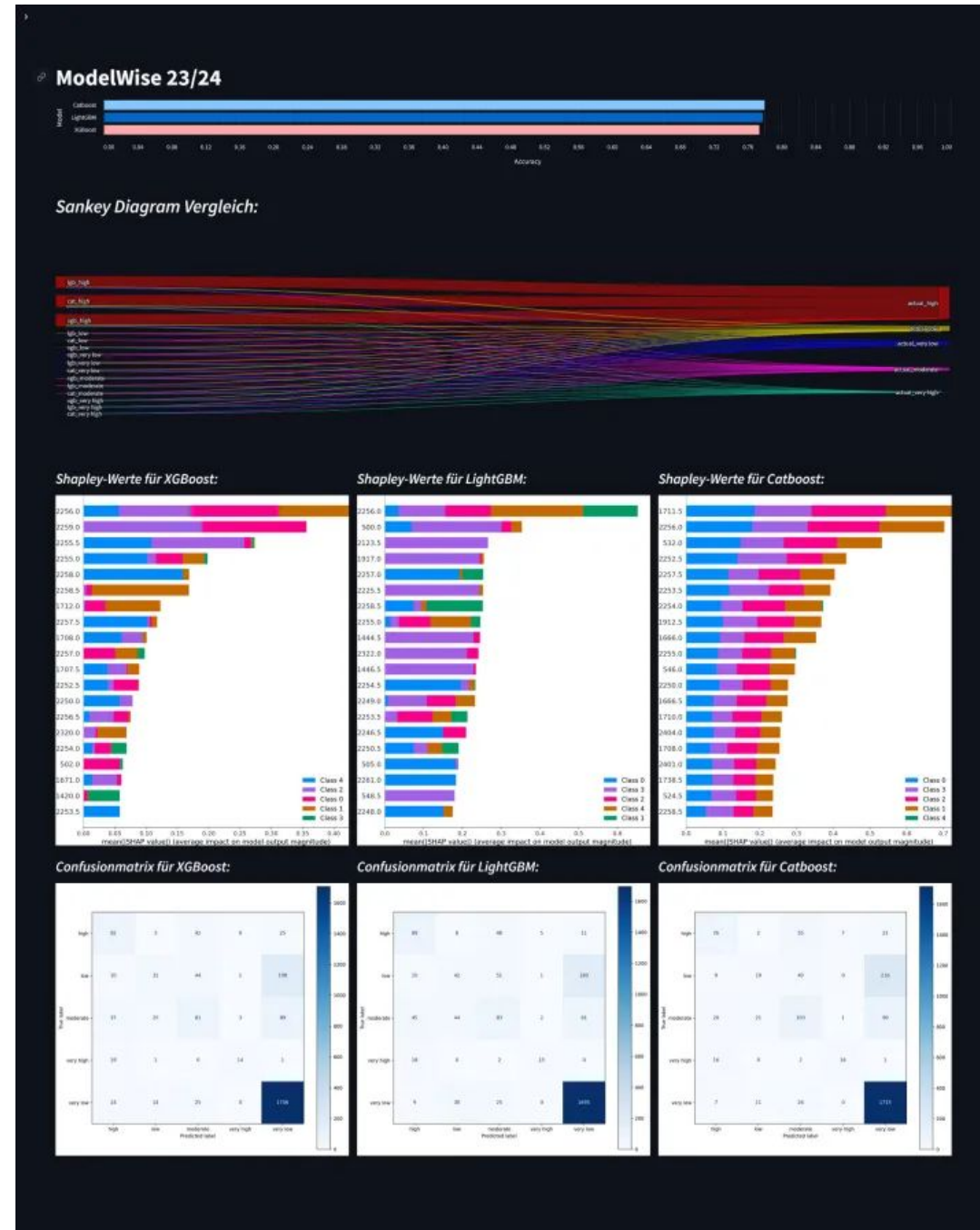
Choose Dataset

- Datasets can be chosen individually
- Leads to better understanding of subsets
- Finetuning in control of user



Compare 3 models

- Focus of our application
- confusion Matrix, shapley-values and accuracy-score
- Interactive sankey-diagram
- Which model is best for our (current) dataset



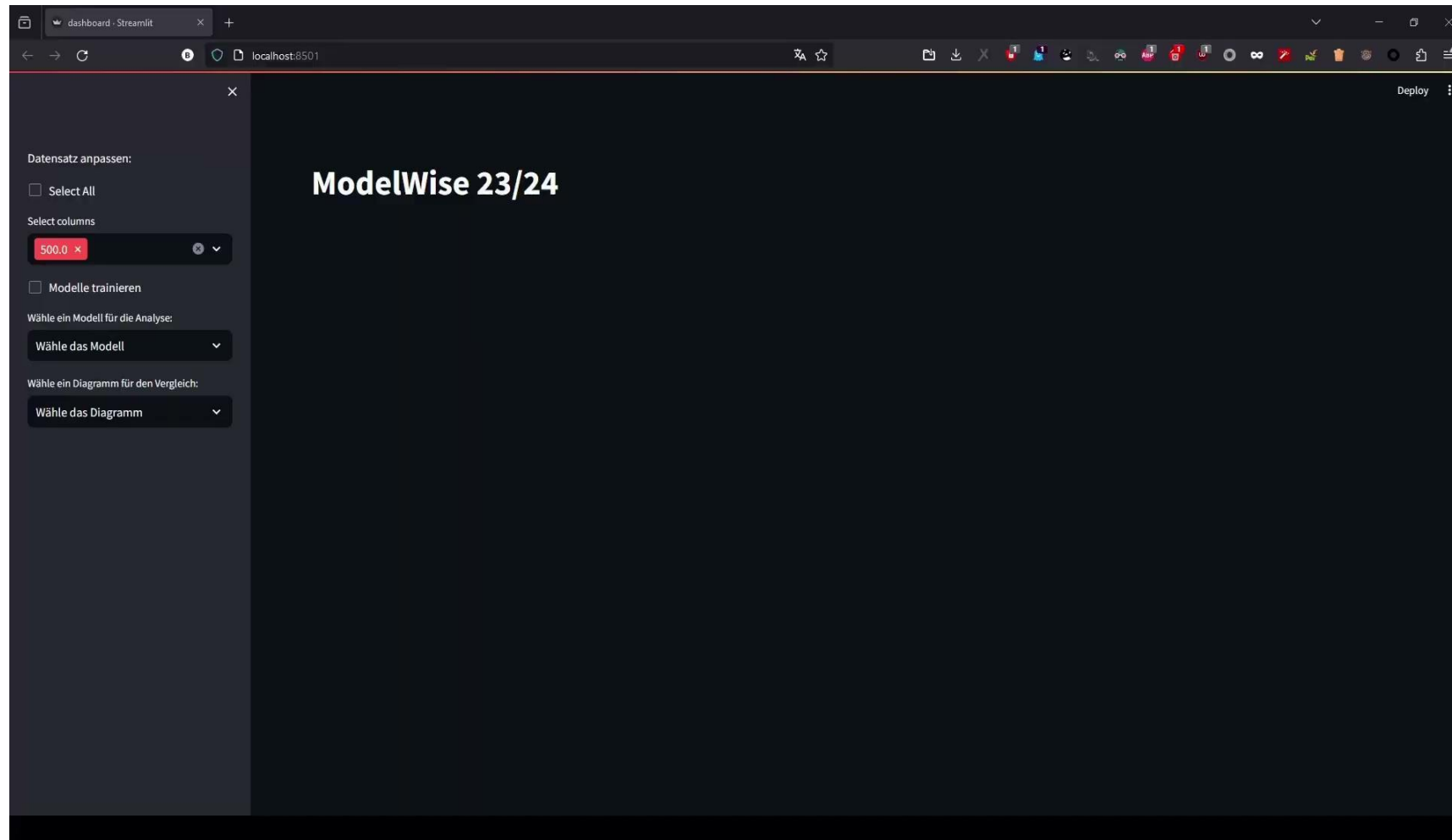
Choose Dataset and compare

Find best model for current dataset

Perhaps different datasets lead to different rankings of models



Demo



Live Demonstration

Conclusion

Pros:

- Lots of comparisons between models and datasets
- Ability to choose every combination of features
- Simple and intuitive design
- Open-Source-Project

Cons:

- No direct exploration of datasets or features (boxplots, scatterplots, etc.)
- Can't download/cache model
- No optimal runtime optimization

Insights

- Biggest challenges:
 - stop adding more stuff and focusing on what we really need (and make it bug-free)
 - making the application user friendly
- What we learned:
 - Streamlit
 - Sankey and shapley
 - Importance of good data
 - Importance of XAI Methods
 - Importance of exploration (of data and models)
 - Using different exploration methods



If you want to know the
future, look at the past.

Albert Einstein

 quote fancy